

Question no 1:

Design a network using **2 switches, 1 router, and 16 PCs**, ensuring that PCs connected to different switches can communicate with each other. Assign appropriate IP addresses and configure the router to act as a gateway between the two subnets.

Instructions:

1. IP Address Allocation:

For **Subnet 1** (connected to Switch 1), configure each PC with IP addresses in the range **192.168.<your registration number>.X** (e.g., if your registration number is 76, use 192.168.76.X for Subnet 1).

For **Subnet 2** (connected to Switch 2), assign IP addresses following **192.168.<next registration number>.X** (e.g., if your registration number is 76, Subnet 2 would use 192.168.77.X).

2. Network Connection Setup:

Connect 8 PCs to each switch.

Link both switches to the router, configuring it to route traffic between the two subnets.

3. Testing:

Verify connectivity by using the ping command.

Test each PC's ability to communicate with all other 15 PCs to ensure complete network functionality.

4. Report Submission:

Submit a step-by-step report of the network setup process.

Include screenshots for each connectivity test, showing successful ping results for each PC tested with all other 15 PCs.

Ensure screenshots clearly demonstrate that all 16 PCs are able to communicate properly.